

Oliver H.

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PROFILE

Computer Science graduate with hands-on robotics experience across ROS2, robotic teleoperation, software-hardware integration, embedded systems, simulation, testing, and debugging. Built and validated ROS2-based robotic systems using Raspberry Pi, microcontrollers, custom controllers, filtering algorithms, and physical robotic platforms. Strong interest in autonomous mobile robots, navigation, perception, sensor integration, and deploying reliable robotics software in real-world environments.

SKILLS

Robotics & ROS : ROS2 Humble, ROS nodes, topics, messages, launch files, RViz, Gazebo Classic, sensor integration, robotic teleoperation, mobile robotics.

Navigation & Localisation : Kalman filtering, particle filters, LiDAR, FastSLAM, probabilistic localisation and path planning.

Programming : Python, C++, C, Java, Ubuntu-based development, object-oriented programming, debugging.

Robotic Systems : Raspberry Pi, microcontrollers, actuators, encoders, custom controllers, software-hardware integration, system validation.

Design & Tools : Git/GitHub, Docker, Linux, Jira, technical documentation, test plans, simulation-based validation.

EXPERIENCE

The University of Edinburgh

Nov. 2025 – Apr. 2026

System Design Project TA

Edinburgh, Scotland

- Mentored **22 engineering teams** building integrated robotic and embedded systems from design through testing and final demonstration.
- Guided debugging of embedded **C/C++/Arduino** control logic, sensors, actuators, ROS node message communication, and software-hardware integration.
- Reviewed technical reports, test plans, and design decisions for clarity, traceability, reproducibility, and correctness.

PROJECTS

ROS2 Robotic Teleoperation Software System | ROS2 Humble, Python, Raspberry Pi, microcontrollers, RViz

- Built and validated a ROS2 dual-arm teleoperation system using custom physical controllers, Raspberry Pi computation, microcontrollers, and robotic manipulators.
- Implemented low-pass and Kalman filtering to reduce artificial tremor while preserving responsive user control.
- Validated system behaviour through latency measurement, RViz visualisation, physical robot testing, failure-case analysis, and technical documentation.
- Winner of the **Skyscanner Accessibility and Inclusion Award 2026**.

Autonomous Pool-Playing Robot | C/Python, microcontrollers, sensors, actuators

- Led low-level implementation and hardware integration for an autonomous pool-playing robot built by an **8-person** engineering team over **8 weeks**.
- Developed and debugged motion-control logic for an XY gantry system, integrating stepper motors, sensors, and coordinate-based actuation.
- Achieved repeatable endpoint positioning within 1–2 cm across a 0.98 m^2 robotic workspace through motion-control calibration and iterative testing.
- Winner of the **Best Technical Design** award 2025 among **16 teams**.

EDUCATION

The University of Edinburgh

Sept. 2021 – May 2026

BEng Computer Science, Upper Second-Class Honours

Edinburgh, Scotland